

Handling Events



Dan Wahlin

Wahlin Consulting

@DanWahlin | www.codewithdan.com



Overview



Instantiating Delegates and Handling Events

Using Delegate Inference



Instantiating Delegates and Handling Events



Event Handler

EventArgs

Event Raiser

Delegate



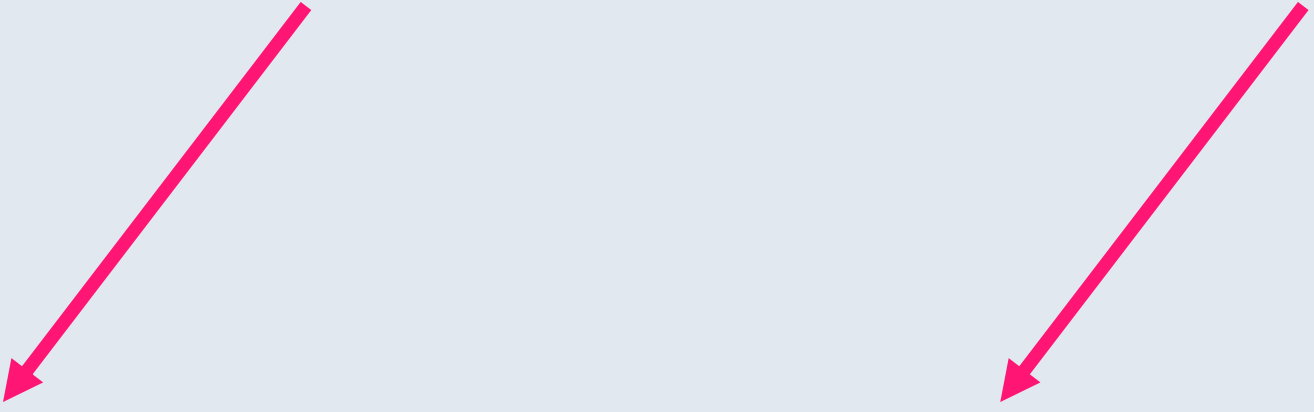


Event Handler

Delegate

```
public delegate int WorkPerformedHandler(object sender, WorkPerformedEventArgs e);
```

```
public void Manager_WorkPerformed(object sender, WorkPerformedEventArgs e) {  
    ...  
}
```



Delegate and Event Handler Method Parameters

The delegate and event handler parameter signatures must match.



```
var worker = new Worker();
```

```
worker.WorkPerformed +=  
    new EventHandler<WorkPerformedEventArgs>(Worker_WorkPerformed);
```

```
void Worker_WorkPerformed(object sender, WorkPerformedEventArgs e)  
{  
    Console.WriteLine(e.Hours.ToString());  
}
```

Defining and Attaching Event Handlers

The **+=** operator is used to attach an event to an event handler.





Delegate Inference




```
var worker = new Worker();
```

```
worker.WorkPerformed += Worker_WorkPerformed;
```

```
void Worker_WorkPerformed(object sender, WorkPerformedHandler e)
{
    Console.WriteLine(e.);
}
```

Delegate Inference

Delegate inference can be used to simplify attaching the event to the event handler.



Summary



The signature of a delegate and event handler must match

To listen to an event, you must construct a delegate and add it to the invocation list

The C# compiler provides “delegate inference” functionality



Up Next:

Lambdas, Action<T>, and Func<T, TResult>

